

Himendra Perera [US Citizen]

khimendraperera@gmail.com – <https://khperera.github.io/Personal-Website/> – (864)-497-2179

Experience

PhD Candidate

August 2019 – Present

Chemical Engineering Department, NC State University

Raleigh, NC

Materials development

- Engineered custom sensor arrays for real-time analysis of force and position, enhancing process feedback frequency and improving process fidelity by 1000%.
- Specialized in imaging technologies, including 3D confocal microscopy and CT imaging, for precisely characterizing bacteria proliferation and suppression in active formulations.
- Conducted benchtop experiments using SolidWorks-designed 3D-printed and CNC-machined parts to develop and validate proof-of-concept mechanical designs and custom analysis techniques.

Integrated software/ML

- Developed advanced statistical and machine learning techniques to design and analyze experiments efficiently.
- Utilized YOLO and fast-RCNN algorithms in PyTorch to identify cells and bio-emulsions, for high-speed, accurate (99.5%) bacteria and particle size analysis.
- Designed optimization algorithms in TensorFlow and Python to optimize and reduce operating costs of supercritical CO₂ operations at Eastman by 20%.
- Analyzed drug delivery rates of different formulations using image processing and diffusion techniques.

Research Engineer

January 2017 – August 2019

Center for Advanced Engineering Fibers and Films

Clemson, SC

- Developed C++ software for motion control of production polymer processing machinery.
- Optimized heat management at interfaces of nanocomposites and biological systems.
- Characterized novel nanocomposite materials for use in medical prosthetics, automotive, and aerospace applications.

Research & Development Engineer

May 2018 – August 2018

Sonoco Products Company

Hartsville, SC

- Led two R&D projects in packaging, collaborating with operators and technicians to develop technology that reduced plastic usage by 50%, saving upwards of \$10 million a year.
- Collaborated with customers in business and medical domains as well as global VPs, aligning customer priorities with strategic business objectives.
- Developed an inline extruder-reactor to increase quality and safety of refeed polymer in food applications.

Research & Development Engineer

May 2017 – August 2017

Sealed Air Corporation

Spartanburg, SC

- Developed novel mushroom-based packaging to replace Styrofoam.
- Utilized ASTM standards and custom tests to measure performance between industry standards and mycelium-based packaging.

Publications

1. **Perera, H.**, Black, B., Ryu, J.E., Corder, R., Khan, S., Rheological Behavior and Roll Coating Properties of PDMS Enhanced with Multi-Walled Carbon Nanotubes and Fumed Silica. *ACS Appl. Eng. Mater.* **2024**, 2, 3, 618–627.
2. **Perera, H.**, Waheibi, R., Sohail M., Khan, S., Hsiao, L., Synthetic Datasets and Machine Learning for Robust Particle Tracking and Intuitive Segmentation in Soft Matter Microscopy. [draft available]
3. **Perera, H.**, Khan, S., Viscoelastic Stresses in Formation and Shaping of Melt Spun Hollow Fibers. [under review]
4. **Perera, H.**, Sohail M., Waheibi, R., Hsiao, L., Khan, S., Confocal Rheo-imaging of Small and Large Amplitude Oscillatory Strain Analysis for Rheological and Microstructural Characterization of Cellulose Acetate Stabilized Pickering Emulsions. [under review]
5. **Perera, H.**, Rahmanian, Khan, S., A data-driven approach for predicting the effect of ScCO₂ on the diffusive properties of polystyrene melt. [draft available]
6. **Perera, H.**, Khan, S., Hsiao, L. Using high frequency rheology to measure fluid properties at high throughput. [in preparation]
7. Islam, M. D., **Perera, H.**, Black, B., Phillips, M., Chen, M.-J., Hodges, G., Jackman, A., Liu, Y., Kim, C.-J., Zikry, M., Khan, S., Zhu, Y., Pankow, M., Ryu, J. E., Template-Free Scalable Fabrication of Linearly Periodic Microstructures by Controlling Ribbing Defects Phenomenon in Forward Roll Coating for Multifunctional Applications. *Adv. Mater. Interfaces* **2022**, 9, 2201237.
8. Liu, S.; Sui, C.; Harbinson, M.; Pudlo, M.; **Perera, H.**; Zhang, Z.; Liu, R.; Ku, Z.; Islam, M. D.; Liu, Y.; Wu, R.; Zhu, Y.; Genzer, J.; Khan, S. A.; Hsu, P.-C.; Ryu, J. E. A Scalable Microstructure Photonic Coating Fabricated by Roll-to-Roll “Defects” for Daytime Subambient Passive Radiative Cooling. *Nano Lett.* **2023**, 23 (17), 7767–7774.
9. Black, B.; Chockalingam, S.; Islam, M. D.; Liu, S.; **Perera, H.**; Khan, S.; Ryu, J. E. Fabrication of Bioinspired Micro/Nano-Textured Surfaces Through Scalable Roll Coating Manufacturing. *Journal of Micro and Nano-Manufacturing* **2023**, 10 (021006).

Education

PhD Biomolecular Engineering (Mechanical Focused) 3.89/4.0

NC State University, Raleigh, NC

B.S. Chemical Engineering 3.98/4.0

Clemson University

Skills

- Programming: Python, Java, C++, C, SQL, JavaScript, HTML, Linux, Git, Signal Processing
- Data science: Pandas, TensorFlow, PyTorch, DeepXDE, SciKit-learn, OpenCV, NumPy, LLM
- Analytical skills: Rheology, SEM, TOF-SIMS, FTIR, DSC, DMA, confocal microscopy, TGA, UV spectroscopy, QCM, AFM, XPS, ellipsometry, nano CT, oscilloscope, 3D scanning, surface profilometry
- Engineering Software: ANSYS, OpenFOAM, SOLIDWORKS, Arduino, Raspberry PI, Blender

References

1. Dr. Saad Khan (academic advisor) khan@ncsu.edu
2. Dr. Lilian Hsiao (committee member) lilian_hsiao@ncsu.edu
3. Dr. Jan Genzer (committee member) jgenzer@ncsu.edu
4. Dr. Ria Corder (former lab member and mentor) riadcorder@gmail.com
5. James Savino (software engineer manager and collaborator) james.savino@homesolutions.com